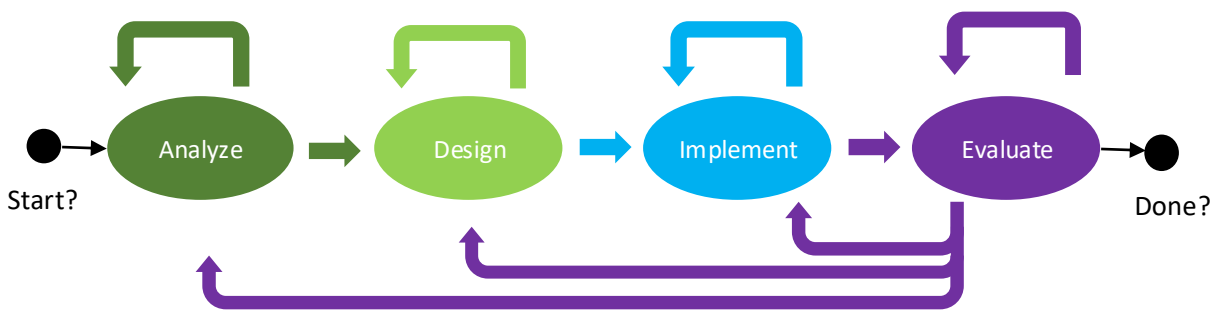
Definitions

Human-centered design (ISO 9241-210)

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- So far, we have discussed how we
  - can analyze problem scenarios (interviews, contextual inquiry, etc.)
  - can generate ideas and quickly communicate them (brainstorming, sketching)
  - can develop prototypes of different degree of fidelity
  - The implementation phase is not covered (software engineering, etc. part of other lectures)
  - But, how can we evaluate our products/prototypes?



Game Interfaces
Evaluation

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Why evaluation?

- It is central part of the human-centered design process
- Identify strengths/weaknesses of designs/interfaces
- Choose one out of multiple design approaches
- Validate design decisions with respect to specified requirements

What to evaluate?

- Compliance to design guidelines, standards, etc.
- Human-machine interaction performance and behavior (productivity, safety, situation awareness, etc.)
- Psychophysiological concepts (stress, workload, cognitive processes, drowsiness, motion sickness, etc.)
- Subjective concepts (User Experience, usability, acceptance, comfort, trust, emotions, wellbeing, etc.)

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Types of evaluation

- Categorization based on timing during the design process
  - Formative evaluation: tests during development, aims at improving in early phases (paper prototypes, wireframes, etc.)
  - Summative evaluation: tests of developed products/features
- Categorization based on utilized methods
  - Analytic evaluation
    - Usability experts, standards, guidelines inspection, literature review, model-based evaluation, cognitive walkthrough, heuristic evaluation
  - Empirical evaluation/User Studies
    - Quantitative: measurement of quantifiable parameters (reaction times, surveys, etc.)
    - Qualitative: Interviews, observation, diary studies, think aloud, or other qualitative methods

Game Interfaces
Definitions and Measurements

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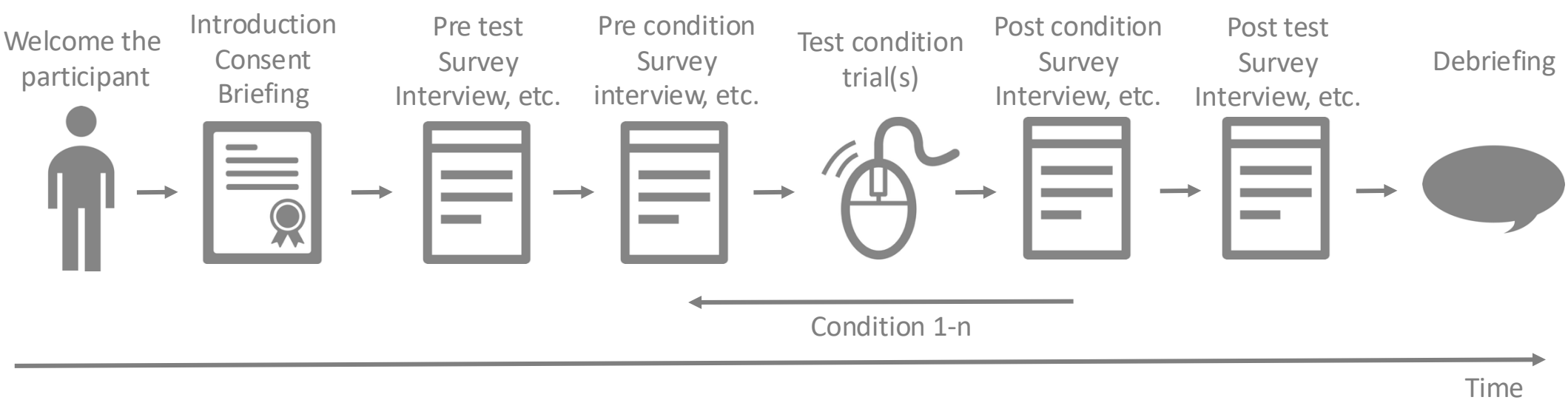
What do we want to assess?

- Human-machine interaction performance and behavior
  - Productivity
  - Safety
  - Situation awareness, etc.
- Psychophysiological concepts
  - Stress
  - Workload
  - Cognitive processes
  - Drowsiness
  - Motion sickness, etc.
- Subjective concepts
  - User Experience
  - Usability
  - Acceptance
  - Trust
  - Comfort
  - Emotions
  - Wellbeing, etc.

Game Interfaces
User Studies Overview

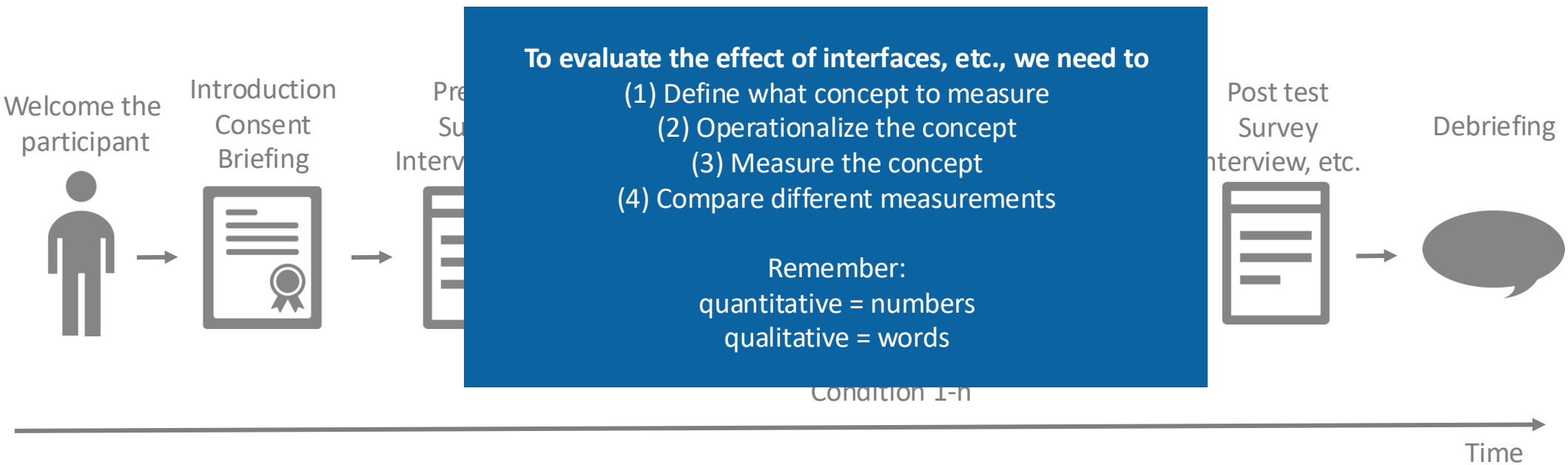
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How a user study looks like



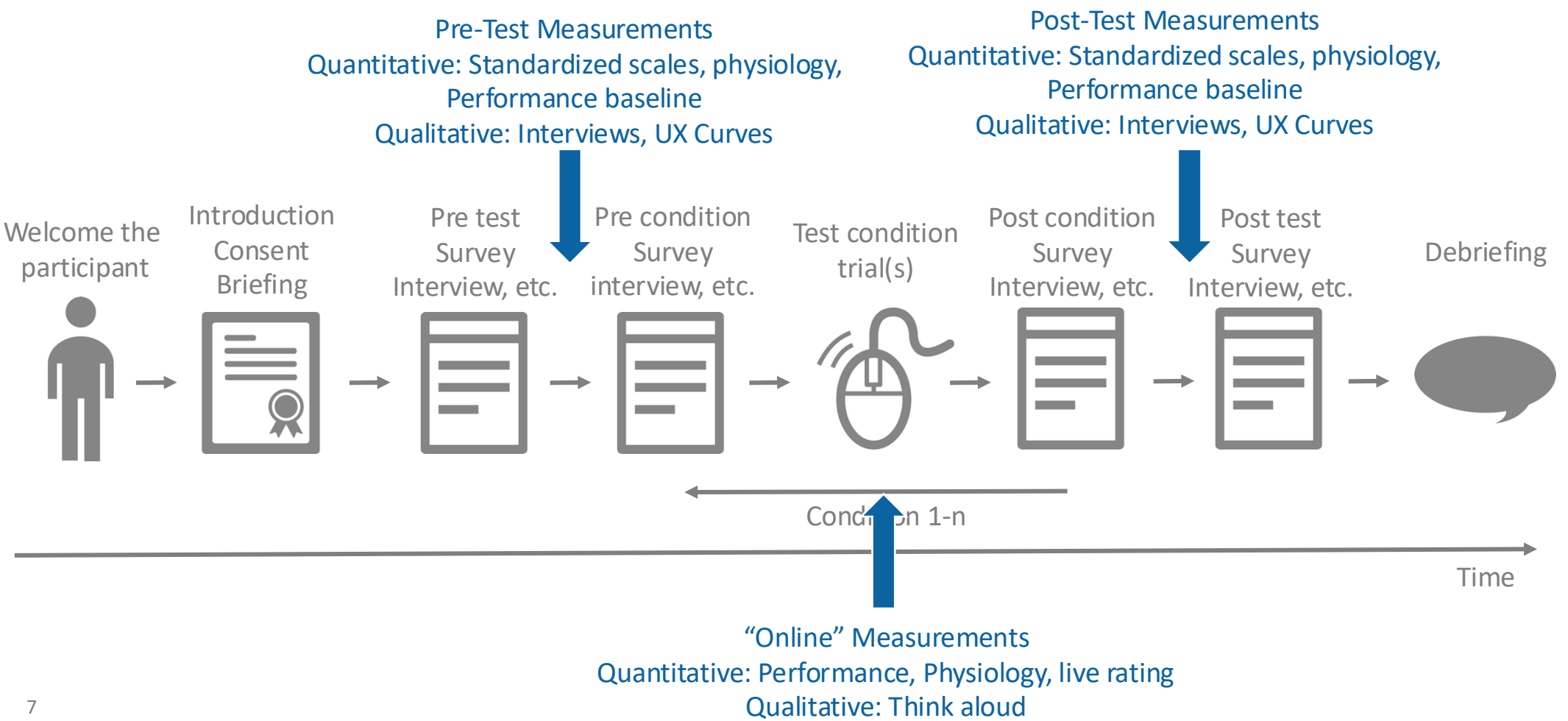
Game Interfaces
User Studies Overview

How a user study looks like



Game Interfaces

User Studies: Measurements



Game Interfaces
Online Measurements

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“Online” Measurements

› Online in this context does not mean “on the web”, but “live” during interaction

Human-machine interaction performance and behavior

› Quantitative:
  › Performance and Behavior: how / how good can users interact with the system?
  › Physiology: What are the effects of the interaction on users’ physiological states?
  › Live Ratings: How do users rate the system while interacting (usually multiple times)

› Qualitative:
  › Think aloud: What do users think about the system while they interact with it?

Many tasks have standardized measurements, but sometimes we have to invent our own!

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Performance Measurements

- Effectiveness: task completion rates, error rates
- Efficiency: time (or other resources) needed to complete tasks
- Depending on concrete task/experiment
  - Texting, Pointing, “Productivity”; or other forms of input (see Lecture on Input): characters per second, Fitts’ law performance, reaction times, etc.

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Performance Measurements: sometimes we have to invent something

› Imagine we want to measure the “effort to change gear in an RPG”

› what could we use?

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Performance Measurements: sometimes we have to invent something

- Imagine we want to measure the “effort to change gear in an RPG”
- what could we use?
- Let the user change gears n=5 (10, 20, …) time
  - Effectiveness
    - Completion rate: Nr. of times the user successfully changed gear / n
    - Error rate: Nr. of times the user failed in completing the task / n
  - Efficiency
    - (Average) time needed to successfully change gear
    - Number of clicks needed to successfully complete the task

Game Interfaces
Quantitative Online Measurements

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Behavioral Measurements vs. Performance

- Sometimes, we do not assess performance but behavior in general
- Example: Use/Disuse, when/where/how often is a feature/product used?
  - Gaming: Playing time in general, in particular game modes, as particular character, use of a weapon etc.
  - Automated driving: user can activate/deactivate automation whenever they want. How long did they use automation/manual driving?
  - Texting: how often did the user select a word suggestion while typing a message?
  - Desktop Application behavior: mouse usage vs. keyboard shortcuts, etc.
  - Mobile Application behavior: In which situations (time, workplace vs. office, etc.) did the user open the app?
  - How often/long did a user look at certain areas in the application (eye-tracking)
  - …and many more

Before doing a test, search the internet for a similar scenario and apply existing measurements (increases comparability), if nothing exists, define something useful!

Game Interfaces
Quantitative Online Measurements

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Physiological Measurements

- Investigate the internal "state" of the user
- Stress measurements (see Lecture on input)
  - Heart rate / heart rate variability (ECG)
  - Galvanic skin response (GSR)
  - Pupil dilatation, etc. (Eye Tracking)
- A collection of different parameters have been used to measure all kinds of concepts like workload, wellbeing, emotions, etc.
- This topic could fill its own lecture…

Before doing a test, search the internet for a similar scenario and apply existing measurements (increases comparability), if nothing exists, define something useful!

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Live Ratings (Discrete)

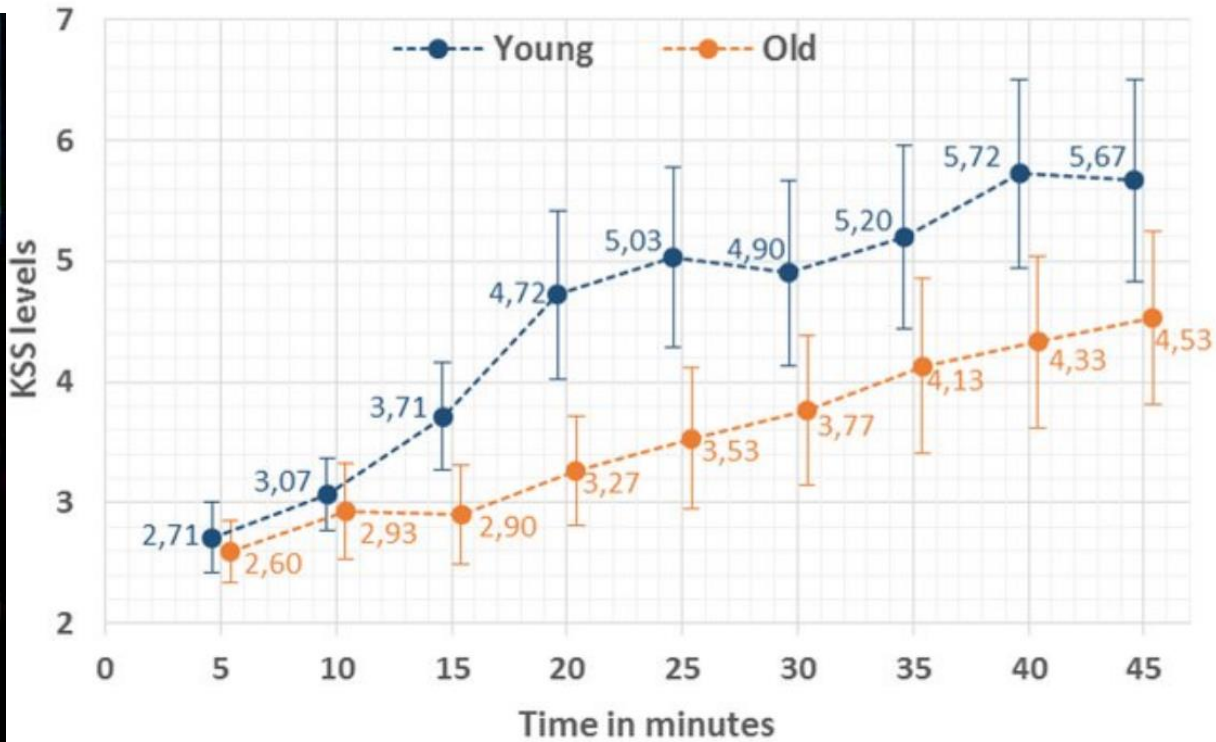
- Users rate some interaction multiple times while interacting
- Typically after some “event” or “frequency”
  - After using a particular tool in a game (“How do you like playing with X?”)
  - After each “Take-Over” in a driving study (“How ready did you feel to Take-Over the car?”)
  - Every 5 minutes in a “fatigue driving” study (“How tired to you feel?”)
  - 6 times a day in a diary study (“How happy do you currently feel”)

Game Interfaces
Quantitative Online Measurements

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Live Ratings (Discrete): Karolinska Sleepiness Scale



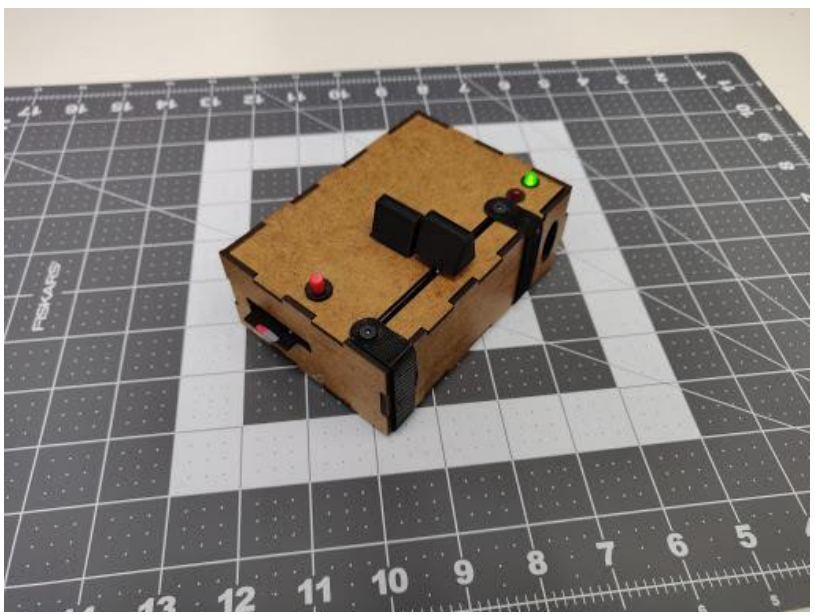


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Live Ratings (Continuous)

› Users rate some concept continuously while interacting (or experiencing)

› “How angry do you feel”, “How do you trust X/Y”, “Rate your experience”, etc.



Slider

Game Interfaces
Post/Pre Test Measurements

Are measurements taken after a test condition is completed

- Quantitative
  - Standardized scales
- Qualitative
  - Interviews
  - Free text fields in surveys
  - (UX) Curves
  - etc.

The boundary between quantitative and qualitative can be fluid in this context!

- Counting the number of statements about “safety” in an interview (quantitative)

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Motivation

- Effectiveness, Efficiency, Performance, etc. can be measured
- But what about subjective factors of the user?
  - Personal preferences
  - Is interacting with the system a satisfying (User) experience?
- Questionnaires help to quantify subjective factors and can provide additional qualitative insights

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Standardized Questionnaires

- Not easy to develop according to scientific criteria
  - Selection and definition of constructs that need to be assessed
  - Formulation and selection of scale items
  - Needs a large sample to validate (N>300)
- Quality criteria
  - Objectivity: Results not depending on experimenter
  - Reliability: Results should be the same when the experiment is repeated
  - Validity: Is the survey assessing exactly what it is designed for?
- In some use cases, it can be sufficient to design an individual survey

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Questionnaires: Additional considerations

- What does it cost?
  - Acquiring participants, phone/Internet costs, pen/paper costs, pilot studies, data management and evaluation, administration, etc.
- What do we want to measure?
  - About the user: Preferences, desires, knowledge, User Experience, acceptance, trust, etc.
  - About the product: Missing features, inconsistent design, behavior in error situations, etc.
- Why is it necessary?
  - Postulate/validate hypotheses, derive research/design roadmap, specify requirements, choose between potential designs, etc.
- What to do with the results?
  - Who will use the data/results, where is it stored, how to deal with missing data/outliers, etc.

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Questionnaires: Major aims

- What questions do we want to ask?
- How long should it take to complete the survey?
- How will the data be analyzed (the desired result must be defined before)?

These considerations will influence content and design of the questionnaire!

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Questionnaires: Content

- Demographics: Age, gender, education, experience, pre-existing knowledge, culture, country of citizenship, etc.
- Main Constructs: User Experience, Acceptance, Workload, etc.
- Design variants: which design is more satisfying, ranking, etc.
- Text fields for additional justifications, opinions, etc.

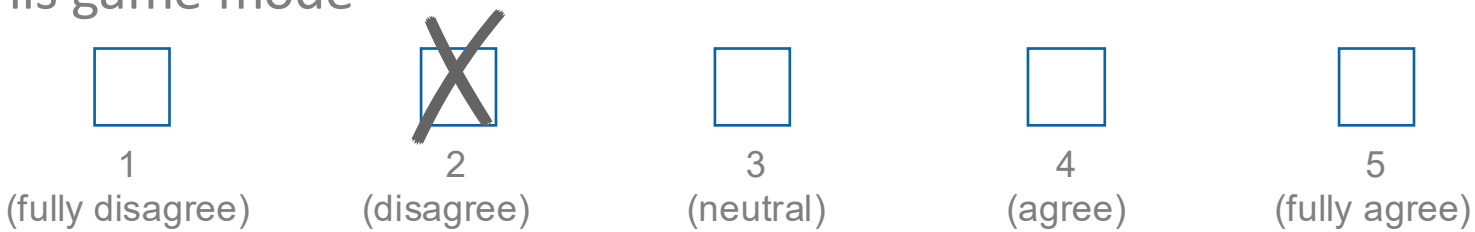
Game Interfaces

Questionnaires – Likert Scale

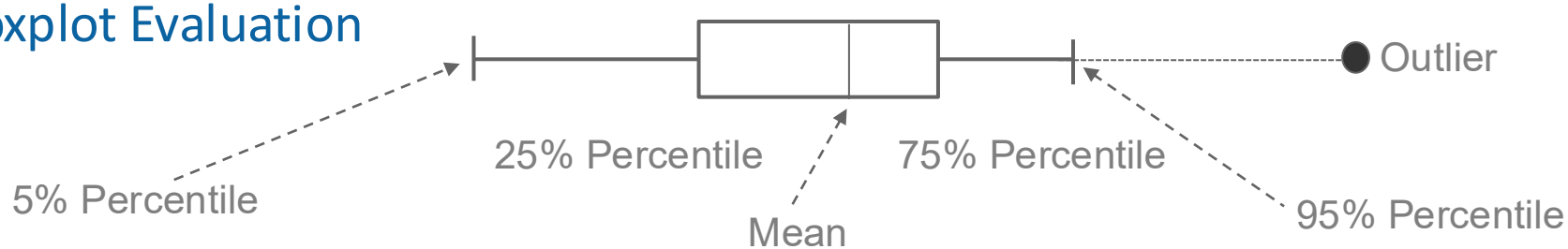
Typically used in psychometric tests, assesses opinions

Agreement rates (with or without neutral element)

"I like this game mode"



Boxplot Evaluation



Game Interfaces
Questionnaires

Experimenter bias

- I better speak in standard German than dialect
  - 2 Interviewers – standard German and dialect
  - Standard German interviewer: 60% agreement
  - Dialect interviewer: 10% agreement

Funding bias

- Participant may give experimenter a satisfying answer
  - How you like X? ✗
  - What is your opinion on X? ?
  - State 3 positive and 3 negative aspects on X. ✓



Game Interfaces
Questionnaires

Priming

› Answers from prior questions influence answers on later questions

A good questionnaire is randomized (order of questions and sections)



What does a cow drink?



Game Interfaces
Post/Pre Test Measurements

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Standardized Scales

- Measure one (or multiple) constructs with scale items
- Are validated and “ready to use”
- Exist for various purposes, for example see https://zis.gesis.org/
- Typically consist of various subscales which “deconstruct” the concept
  - Example: Trust = Trust / Distrust
- A sub-scale may consist of one or multiple items
  - “The system is reliable”
  - “The system is dependable”
  - “I trust the system”
- Subscale calculation: Average, Sum, Max, etc.
- “Overall score”: Some scales can be aggregated into a single score in similar fashion, i.e., Average, Sum, etc. or more complex calculated (weighted, etc.)

Game Interfaces

Game UX (GUESS-18)

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› 9 Dimensions: Usability/Playability; Narratives; Play Engrossment; Enjoyment; Creative Freedom; Audio Aesthetics; Personal Gratification; Social Connectivity; Visual Aesthetics

Game Interfaces

Game User Experience Satisfaction Scale (GUESS-18)

Game UX

Dimensions: Usability, Playability, Narratives, Enjoyment, etc.

Usability/Playability	I find the controls of the game to be straightforward.
	I find the game's interface to be easy to navigate.
Narratives	I am captivated by the game's story from the beginning.
	I enjoy the fantasy or story provided by the game.
Play Engrossment	I feel detached from the outside world while playing the game.
	I do not care to check events that are happening in the real world during the game.
Enjoyment	I think the game is fun.
	I feel bored while playing the game. (REVERSE CODE)
Creative Freedom	I feel the game allows me to be imaginative.
	I feel creative while playing the game.
Audio Aesthetics	I enjoy the sound effects in the game.
	I feel the game's audio (e.g., sound effects, music) enhances my gaming experience.
Personal Gratification	I am very focused on my own performance while playing the game.
	I want to do as well as possible during the game.
Social Connectivity	I find the game supports social interaction (e.g., chat) between players.
	I like to play this game with other players.
Visual Aesthetics	I enjoy the game's graphics.
	I think the game is visually appealing.

Measurement: Again a subjective scale with Likert-scale questions

Keebler, J. R., Shelstad, W. J., Smith, D. C., Chaparro, B. S., & Phan, M. H. (2020). Validation of the GUESS-18: a short version of the Game User Experience Satisfaction Scale (GUESS). *Journal of Usability Studies*, *16*(1), 49.

Game Interfaces

Game UX (GUESS-18)

- 9 Dimensions: Usability/Playability; Narratives; Play Engrossment; Enjoyment; Creative Freedom; Audio Aesthetics; Personal Gratification; Social Connectivity; Visual Aesthetics
- GUESS-18 Scale
  - Reverse scale items if needed (only item „bored“ in GUESS-18)
  - 18 item scale, each rated from 1 to 7 (Likert scale)
  - Average the 2 items in each dimension
  - Composit score: sum of all dimensions (9-63)

Game Interfaces
Workload

Workload “represents the costs of accomplishing [mission] requirements for the human operator [user]” [1]

NASA Task Load Index

- Mental Demand
- Physical Demand
- Temporal Demand
- Effort
- Performance
- Frustration Level

- Item Example: “How hurried or rushed was the pace of the task” (Temporal)

[1] Hart, S. G. (2006, October). NASA-task load index (NASA-TLX); 20 years later. In Proceedings of the human factors and ergonomics society annual meeting (Vol. 50, No. 9, pp. 904-908). Sage CA: Los Angeles, CA: Sage publications.

Game Interfaces

Different Results based on Perspective

› „Steer-By-Wifi: Lateral Vehicle Control for Take-Overs with Nomadic Devices“

Steer-By-WiFi



Clemens Schartmüller, Andreas Riener, Philipp Wintersberger
Technische Hochschule Ingolstadt, Germany

Mensch in Bewegung

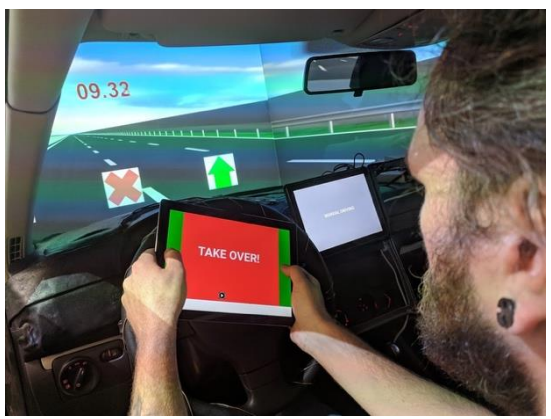
Game Interfaces

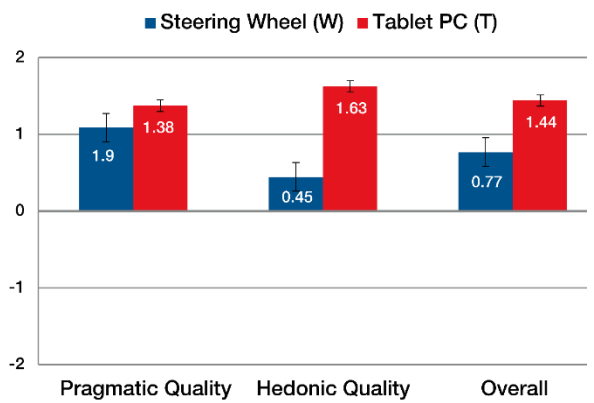
Different Results based on Perspective

› „Steer-By-Wifi: Lateral Vehicle Control for Take-Overs with Nomadic Devices“

- › Better steering performance with the steering wheel
- › Better UX with a tablet as control
- › https://youtu.be/Y34248W5BvI

HF-Expert:
„let‘s better stick to the steering wheel!“





UX-Expert:
„let‘s try out the tablet steering!“

Game Interfaces
Qualitative Evaluation

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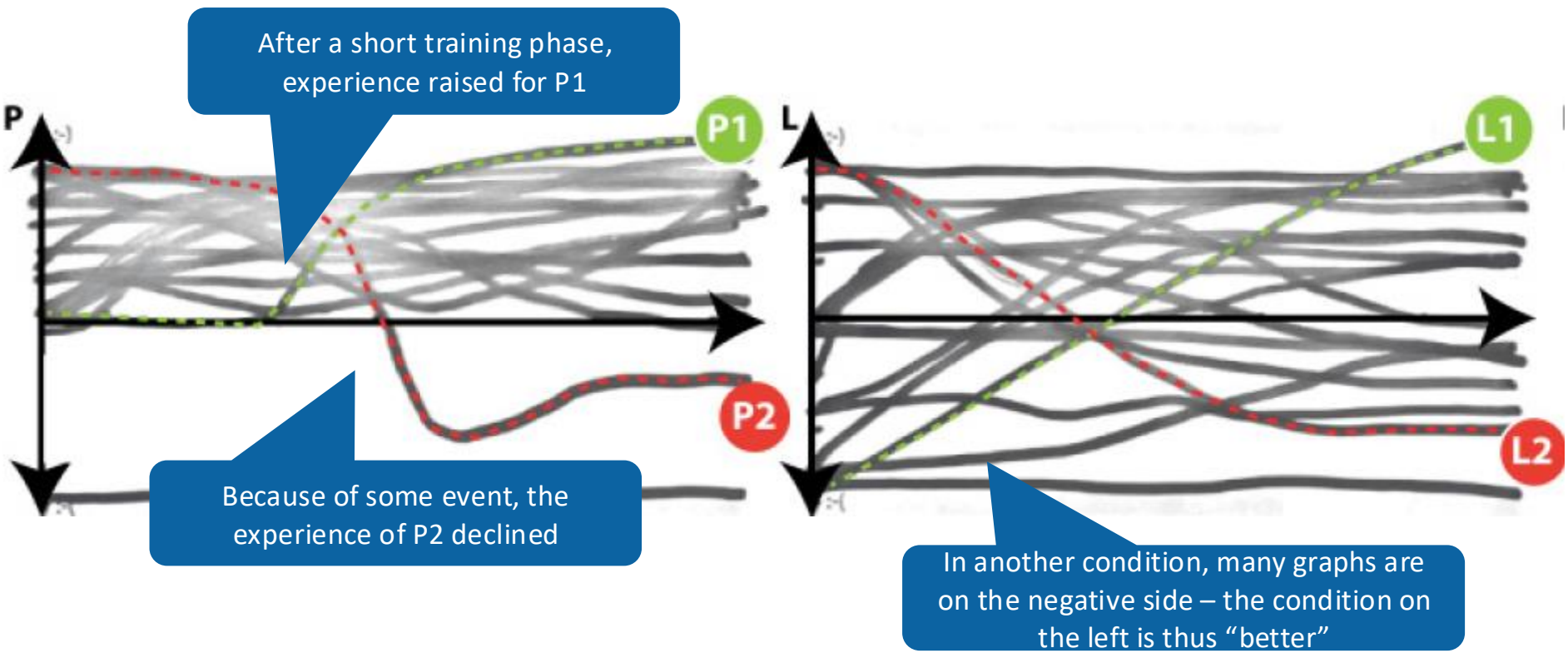
UX Curves

- A tool that is used post-test (retrospective)
- Participants draw a graph in a diagram that represents their feeling(s)
  - The timeline is on the x-axis
  - The “experience” is on the y-axis (+ means better experience, - means lower experience)
- While drawing, participants “think aloud” and express why the curve changes
- These interviews are recorded
- The result can later be analyzed quantitatively (regarding the curves) and qualitatively (regarding the statements)
- UX-Curves are a retrospective version of a “continuous live rating” as shown before
- The method can be used for other constructs: Trust Curve, Emotions, etc.

Game Interfaces
Qualitative Evaluation

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UX Curves Example



Game Interfaces
Evaluation

Experimental Design

- In an experiment, we want to evaluate differences between test conditions
- Typically, we compare two or more test conditions
  - We compare the presence of a feature to the absence of the feature (“baseline”)
  - We compare multiple design variants (with or without baseline)
- Variables
  - Independent variable: represents the conditions
  - Dependent variable: represents the measurement outcome
  - “Pragmatic Quality is dependent on the condition (i.e., design, etc.)”
- Group Design
  - Within-group design: All participants experience both conditions
  - Between-group design: Half/Third of participants experience condition A, the other half B (and a third C), …

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Experimental Design

- **Mixed Design:** involve at least 1 within, and at least 1 between subjects factor
- Example: we want to compare 2 design variants, both in 2 colors (red/blue)
- Each participant experiences 2 conditions, 1 red, 1 blue, but only one variant
- **Between subjects factor:** design variant
- **Within subjects factor:** color
- To reduce order effects, we randomize (p1 starts with blue, p2 with red, …)

	A / B (between group design)	
Red & blue (Within group design)	A (red)	B (red)
	A (blue)	B (blue)

Game Interfaces
Evaluation - Variables

	Data type	Meaning	Example	Allowed Transformations
Non-parameteric	**Nominal data**	Equality/ Unequality	e.g., Gender	All bijective transformations, e.g., if x1≠x2, then y1≠y2
Non-parameteric	**Ordinal data**	More-Less-Relation	e.g., Degree of Education, Rankings	All strictly monotonic transformations, e.g., if x1<x2<x3, then y1<y2<y3 or y1>y2>y3
Parameteric	**Interval scale data**	Equality of Differences	e.g., Temperature	All linear transformations, e.g., y = a * x + b
Parameteric	**Ratio scale data**	Equality of ratios	e.g., Weight, Length	All similarity transformations, e.g., y = a * x

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Basic Terms

- Mean
  - Average of all measurements
  - For a useful calculation, the data must at least be at ordinal scale
- Median
  - The value splitting the data in two equal-sized parts (half above, half below)
  - At the median, the average difference of all other measurements is smallest
- Mode
  - The most frequent value in a discrete distribution
  - Can also be determined for data at ordinal scale

Basic Terms

› Standard Deviation

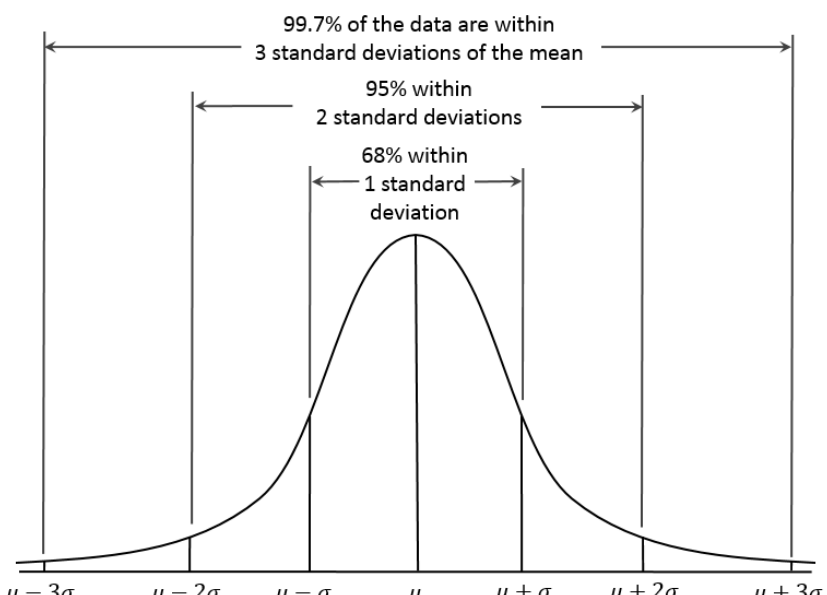
- › Distance of the mean value to the turning point of the normal distribution
- › Shows the scatter in the data

$$\sigma_x = \sqrt{\sigma_x{}^2} = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$

› Variance

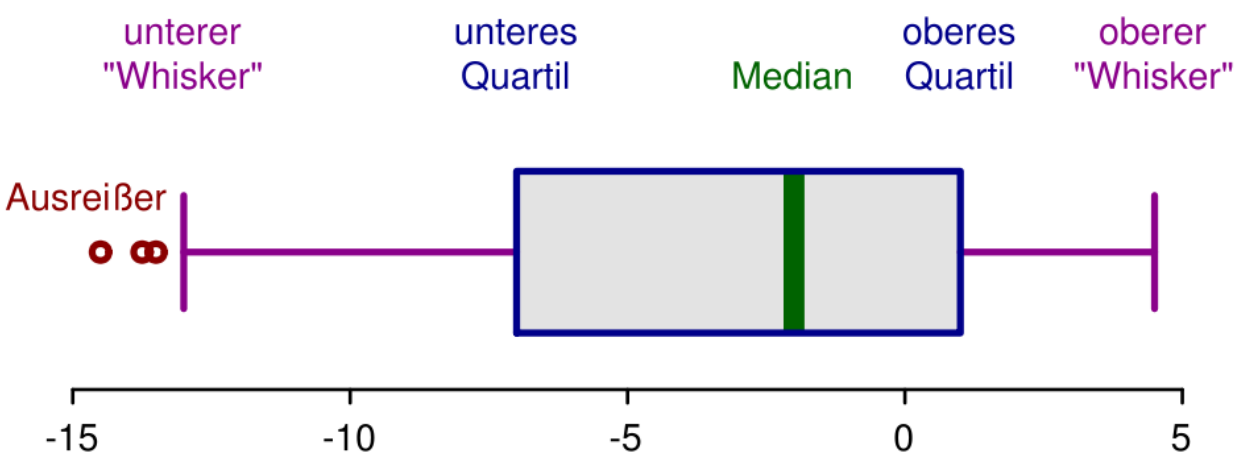
- › Square of the standard deviation

$$\sigma_x{}^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$$



Basic Terms

- Median and Quartiles; Boxplot
  - Median: half of the measurement points is above, the other half below the median
  - Top/Bottom Quartile: 25% of measurements point below/above value
  - “Whisker”: Often minimum and maximum, sometimes without outliers



Game Interfaces
Evaluation

Evaluation Process

- Define experiment (evaluation plan)
- Conduct experiment (record data)
- Preprocessing (bring data in format suitable for evaluation)
- Choose suitable statistical tests
- Evaluate
- Interpret the results
- Reporting

Game Interfaces
Evaluation

Evaluation Process

- Define experiment (evaluation plan)
  - Which constructs should be assessed (i.e., performance, behavior, UX, acceptance, etc.)
  - Which dependent/independent variables?
  - Which experimental design (within group, between groups, mixed)
  - How will the procedure look like?
  - Formulate hypotheses (“design 1 will lead to a significantly higher UX than design 2”)
- Conduct experiment (record data)
- Preprocessing (bring data in format suitable for evaluation)
- Choose suitable statistical tests
- Evaluate
- Interpret the results
- Reporting

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Evaluation Process

- Define experiment (evaluation plan)
- Conduct experiment (record data)
  - Experiment guideline/checklist
  - The process must be the same for each participant
  - Tell participants everything but only what they need to know
  - Always think about potential biases!
- Preprocessing (bring data in format suitable for evaluation)
- Choose suitable statistical tests
- Evaluate
- Interpret the results
- Reporting

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Evaluation Process

- Define experiment (evaluation plan)
- Conduct experiment (record data)
- Preprocessing (bring data in format suitable for evaluation)
  - Example: derive locations from a recorded list of GPS positions
  - i.e., “any set of positions within 100m where a participant was longer than 10mins is considered as a location”
  - everything else counts as movement and will be eliminated
- Choose suitable statistical tests
- Evaluate
- Interpret the results
- Reporting

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Evaluation Process

- Define experiment (evaluation plan)
- Conduct experiment (record data)
- Preprocessing (bring data in format suitable for evaluation)
- Choose suitable statistical tests
- Evaluate
- Interpret the results
- Reporting

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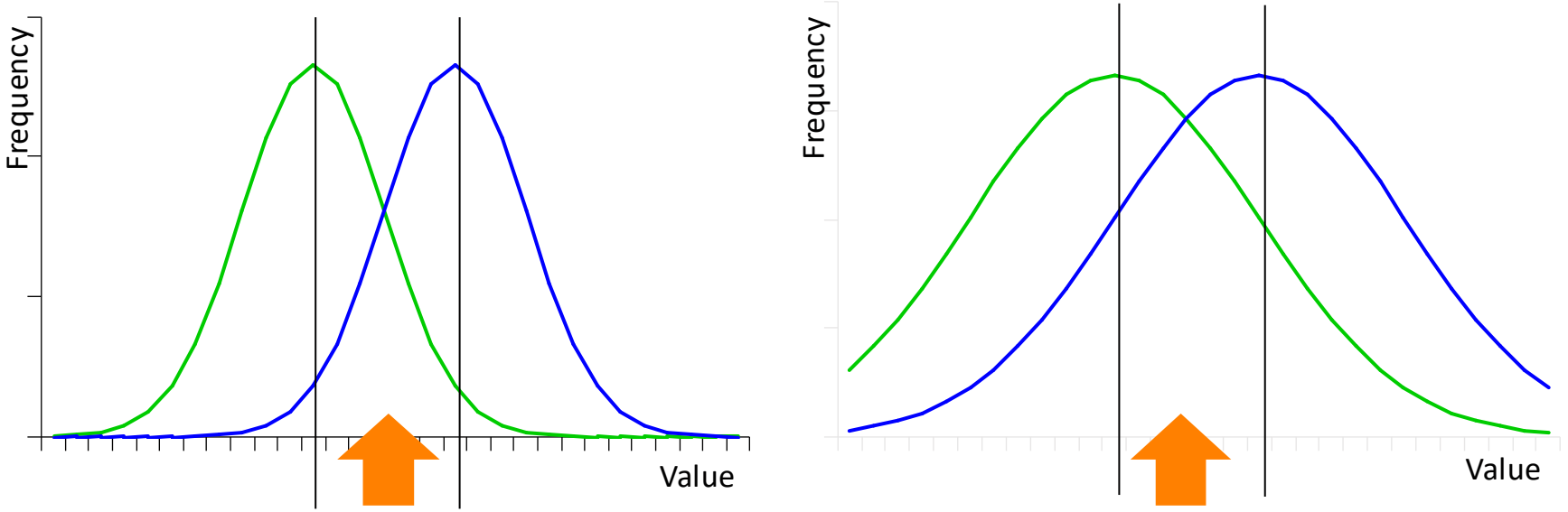
Significance Testing

- The right choice for a test depends on preconditions
  - The number of conditions
  - The type of the dependent variable
  - The experimental design: between vs. within group
  - The distribution of the data (parametric test assume normal distribution)
  - “Central Limit Theorem”: N>30 participants, a normal distribution can be assumed
- Example: Student’s t-test
  - Compares means of two groups
  - Assumes normal distribution

Significance Testing

› **t-test:** examines whether two averages are systematically different

› The result tells us if the difference did not appear “by chance”



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Significance Testing

- Interpretation: the result (p-value) tells you the probability that 2 values come from the same group
  - Hypothesis: the two groups are different
  - Null hypothesis: the two groups are the same
- Interpretation
  - P-value of 0.6: 60% chance that the data belongs to the same group
  - P-value of 0.1: 10% chance that the data belongs to the same group
  - At which P-value you claim a “statistically significant difference” depends on your criteria!
  - Typically, the threshold is $p<.05$
  - In both of the above cases, we reject the Hypothesis: there is no difference between the two groups

Significance Testing

- Interpretation: p-value <.05
  - There is a chance <5% that the data belongs to the same group
  - In this case we reject the Null Hypothesis, and accept the Hypothesis: there is a difference between the two groups
- The “Null Hypothesis”
  - Hypothesis testing is about rejecting the Null Hypothesis
  - When we “fail” to reject the Null Hypothesis, this does not mean we can accept it!
  - In such a situation, we did not find a difference, but
    - What if another sample (selection of participants) would have led to a difference?
    - What if the sample size was too small, the variability in the data to high, or we suffered noisy data?
  - Long story short: when we find a difference it is there, but we cannot assume that there is no difference just because we did not find it!

Game Interfaces
Evaluation

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Choosing the right test

- Parametric tests
  - When the data is normally distributed
  - T-test: 2 conditions (within: paired samples t-test; between: independent samples t-test)
  - Repeated Measurement ANOVA: 3 or more conditions
- Non-Parametric tests
  - When the data is NOT normally distributed
  - Wilcoxon signed rank test: 2 conditions, dependent samples
  - Mann-Whitney-U test: 2 conditions, independent samples
  - Friedman ANOVA: 3 or more conditions, dependent samples
  - Kruskal-Wallis test: 3 or more conditions, independent samples

- Choosing the right test is not trivial!

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Evaluation Process

- Define experiment (evaluation plan)
- Conduct experiment (record data)
- Preprocessing (bring data in format suitable for evaluation)
- Choose suitable statistical tests
- Evaluate
- Interpret the results
  - Relate p-values, sample size, and effect size
  - Interpret with respect to your hypotheses (accept/reject)
  - Discuss your results in relation to other experiments
- Reporting

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Evaluation Process

- Define experiment (evaluation plan)
- Conduct experiment (record data)
- Preprocessing (bring data in format suitable for evaluation)
- Choose suitable statistical tests
- Evaluate
- Interpret the results
- Reporting
  - Document all the steps outlined in this process
  - Report everything important, do not leave things out intentionally
  - Use the correct language and formulas to report statistics (see https://statistics.laerd.com/)

Evaluation – Group Task for Exercises

Game Interfaces
Evaluation of an Interface for Item Selection

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Evaluation Process

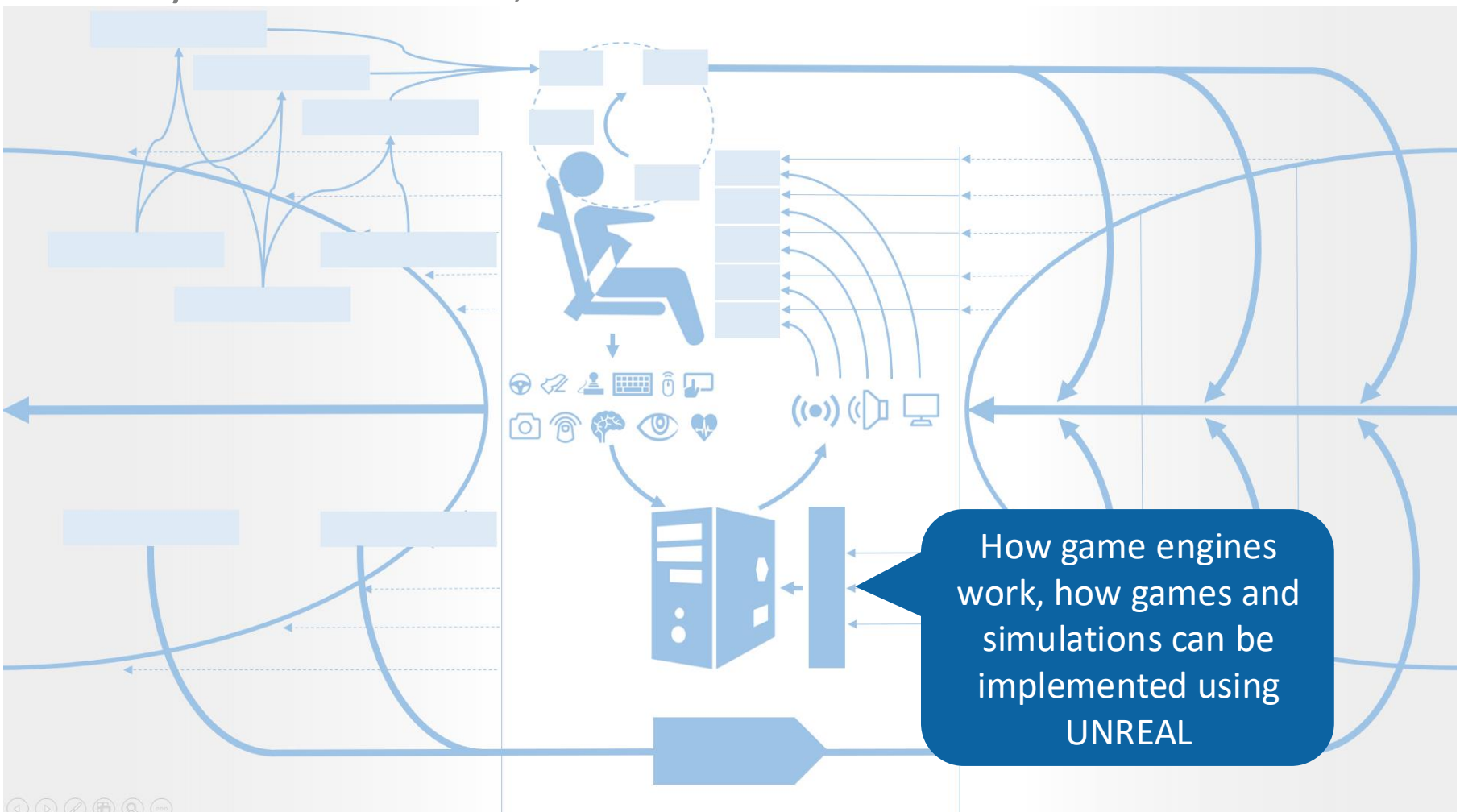
- Define experiment (evaluation plan)
  - Independent Variable(s) i.e., Conditions
  - Dependent Variables
  - Experimental Design (within, between, mixed)
  - Procedure
  - Participants



Module Summary

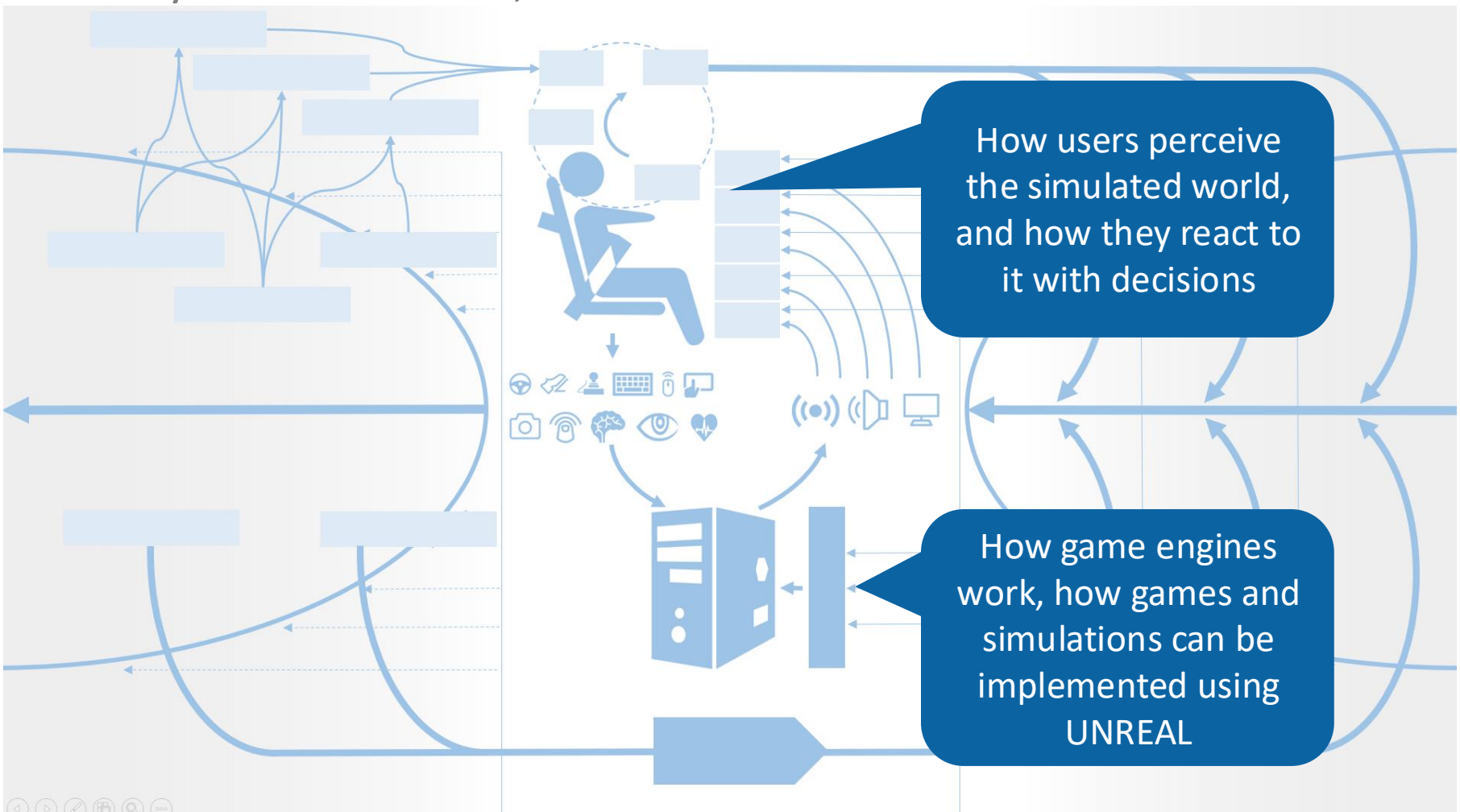
Game Engine Architectures: Evaluation

Summary: In this module, we have learned



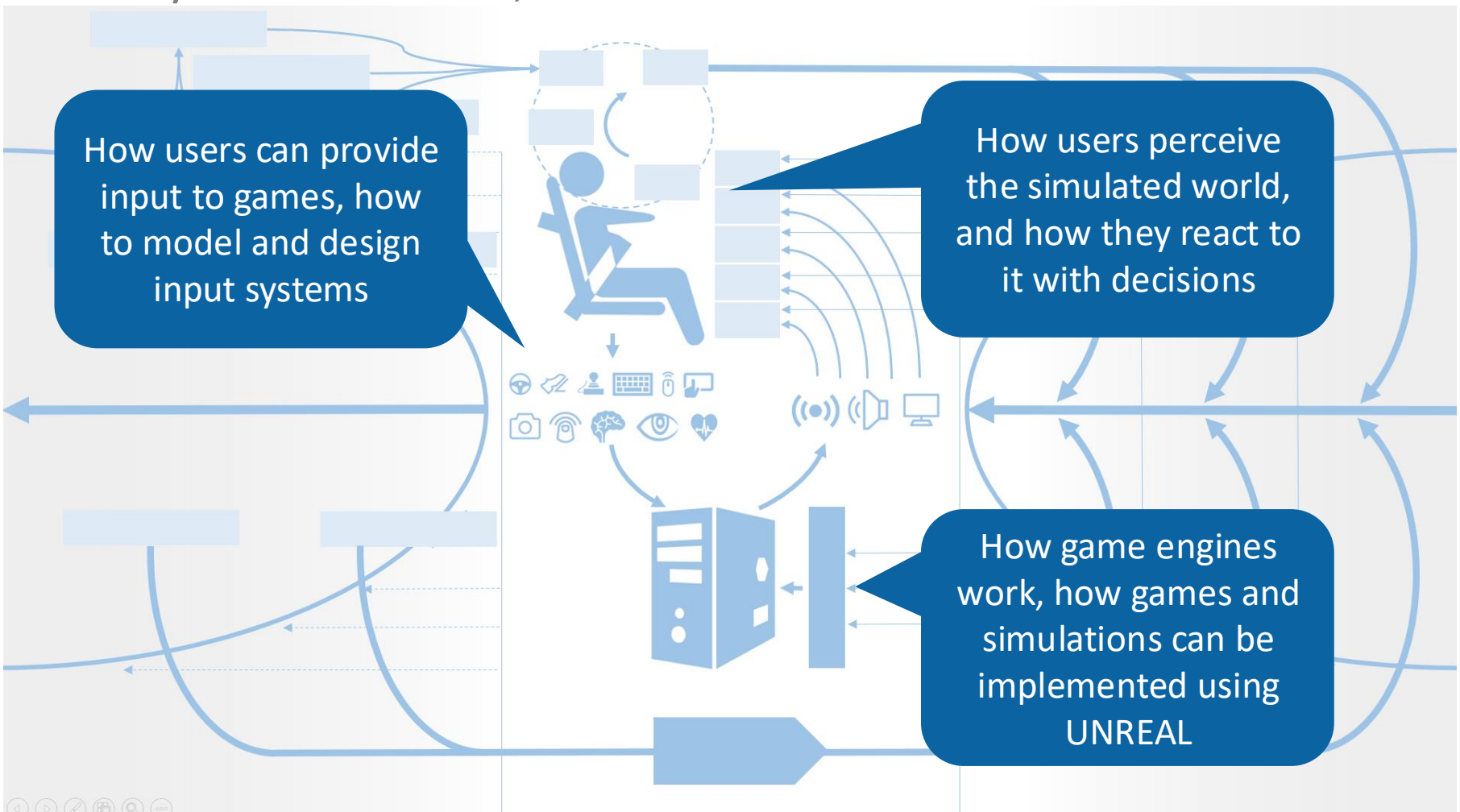
Game Engine Architectures: Evaluation

Summary: In this module, we have learned



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